

Structure & functions of macromolecules – Nucleic Acids: Study Notes

Session Objectives:

SOM.MK.I.BPM1.1.FTM.3.BCHM.0020	Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0021	Describe the phosphodiester bond
SOM.MK.I.BPM1.1.FTM.3.BCHM.0022	Describe the three important features of the DNA double helix (complementary, antiparallel, and forces stabilizing the helix).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0023	Discuss the physicochemical properties of DNA
SOM.MK.I.BPM1.1.FTM.3.BCHM.0024	Describe the basic structures and roles of the major classes of RNA (mRNA, tRNA, rRNA, snRNA, miRNA)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0025	Predict the sequence of the complementary strand if the sequence of one of the DNA strands is provided.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0026	Analyze the nucleotide composition of double stranded DNA using the Chargaff's rule

Recommended Reading: Lippincott's Biochemistry 8th Edition

<http://sgu.idm.oclc.org/login?url=https://meded.lwwhealthlibrary.com/book.aspx?bookid=3073> Chapter 22 (Pages 324 – 325), Chapter 30 (Pages 460 – 462), Chapter 31 (Pages 482 – 484).

IMPORTANT: Please see the Panopto videos on the topic of Nucleic Acids.

Function:

The function of nucleic acids is primarily as informational biomolecules that store and transmit genetic (hereditary) information. They generally occur as large un-branched polymers of nucleotides called RNA and DNA.

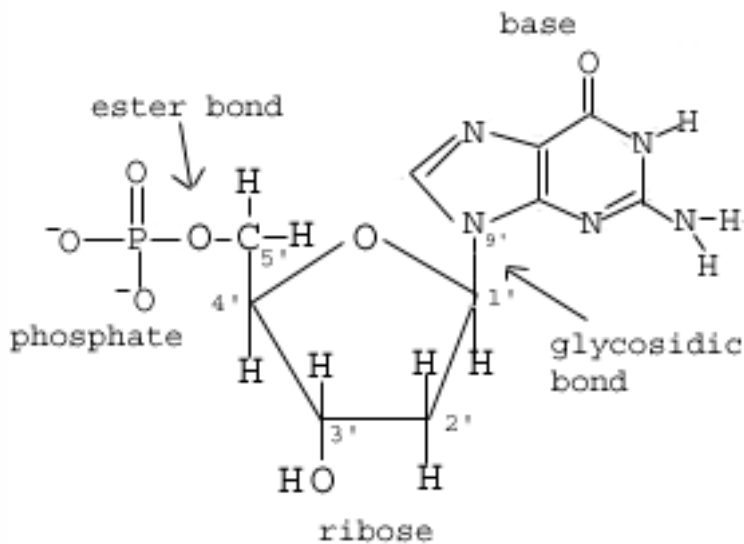
Nucleotides are the monomeric building blocks of all nucleic acid polymers. Each nucleotide consists of a heterocyclic nitrogenous base, either a purine or pyrimidine, a pentose sugar in the form of a ribose (RNA) or deoxyribose (DNA) and a phosphate group.

Nucleosides and Nucleotides:

Nucleoside = base + sugar

Nucleotide = base + sugar + phosphate

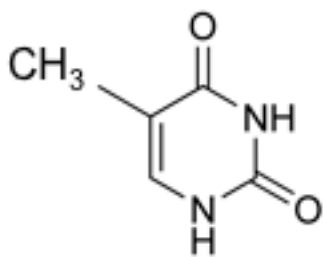
Therefore, nucleotides can be considered nucleoside-mono phosphates – usually 5'-phosphates. The base is linked by a β -N-glycosidic bond to C-1' of the pentose and N-1 of pyrimidines and N-9 of purines. The phosphate group is attached to the C-5' of the pentose by an ester bond. See below an example of the structure of a nucleoside-mono phosphate.



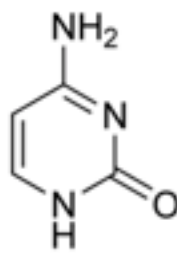
Bases:

The bases found associated with RNA and DNA occur in the form of purines or pyrimidines.

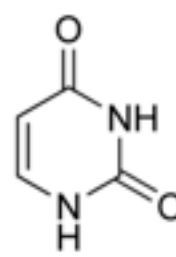
Pyrimidines are heterocyclic compounds containing nitrogen atoms at position 1 and 3 of a 6-member ring. Pyrimidines include thymine, cytosine and uracil.



thymine

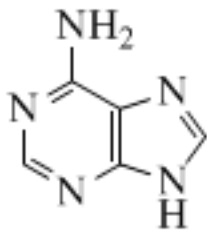


cytosine

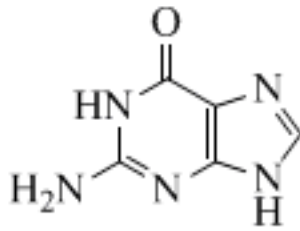


uracil

Purines are heterocyclic compounds consisting of a pyrimidine fuse to a 5 member imidazole ring. Purines include adenine and guanine.



adenine



guanine

The bases associated with a nucleotide that are found in RNA are: adenine, guanine, cytosine, and uracil.

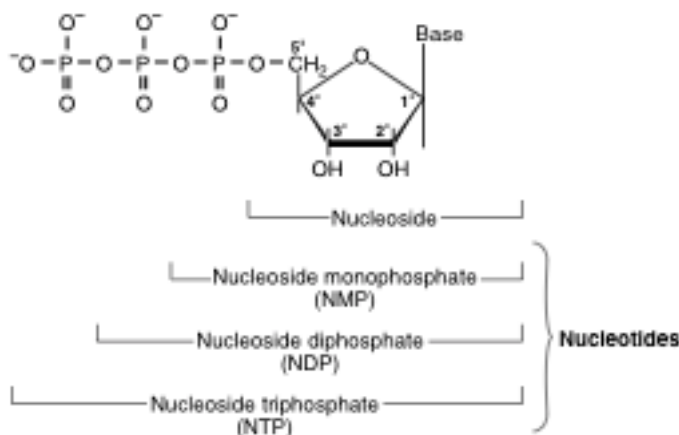
There are several rare bases found within RNA, specifically transfer RNA (tRNA), that are involved in of amino-acyl tRNA synthetase/tRNA recognition, tRNA/codon recognition and peptide synthesis by the ribosome. These rare modified bases are: 4-thiouracil, dihydrouracil, and 6 mercaptopurine.

The bases associated with a nucleotide found in DNA are: adenine, guanine, cytosine, and thymine.

In DNA, the common minor bases are those that have been modified by the addition of a methyl ($-\text{CH}_3$) group. In eukaryotes, these modifications play important roles in regulation of gene transcription and mRNA translation. In viral DNA certain bases may be hydroxymethylated or glycosylated. In bacteria and viruses, these modifications play a role in gene expression and in protection of the DNA from endonuclease digestion. These minor bases include: 5-methyl-C, 5-hydroxymethyl-C, 6-methyl-A, 2-methyl-G, 7-methyl-G.

Nucleoside triphosphates:

Ribonucleoside triphosphates and deoxyribonucleoside triphosphates serve as precursors or substrates for the synthesis of RNA and DNA respectively.



Nucleotides are acids at neutral pH and possess two negative charges on the phosphate. Nucleoside-diphosphates and –triphosphates have 3 and 4 negative charges respectively.

Thus, the following nomenclature has developed (example):

<u>BASE</u>	<u>NUCLEOSIDE</u>	<u>NUCLEOTIDE</u>	<u>ABBREVIATION</u>
Adenine	adenosine	adenylic acid	AMP dAMP
Guanine			
Cytosine			
Thymine			----
Uracil			----

Functions of Nucleoside Triphosphates:

- precursors of the nucleotides for RNA and DNA
- ATP – universal chemical energy of biological systems
- GTP - energy source for protein biosynthesis and involved in signal transduction
- UTP – activation of sugars in biosynthesis of CHO and glycoproteins
- CTP – activation of reactants in phospholipids biosynthesis
- Nucleotide derivatives function as coenzymes. Eg. Pyridine and flavin coenzymes

DNA and RNA Structure:

DNA and RNA are generated by the polymerization of nucleotides to form polynucleotides. The link, in both RNA and DNA, between the nucleotide monomers is a 3', 5' –phosphodiester bond. The polynucleotide chains have polarity: a 5' end defined by the presence of a phosphate group bound to the 5' carbon atom of the sugar, and a 3' end defined by the presence of a hydroxy (-OH) group bound to the 3' carbon atom of the sugar (see structure below).

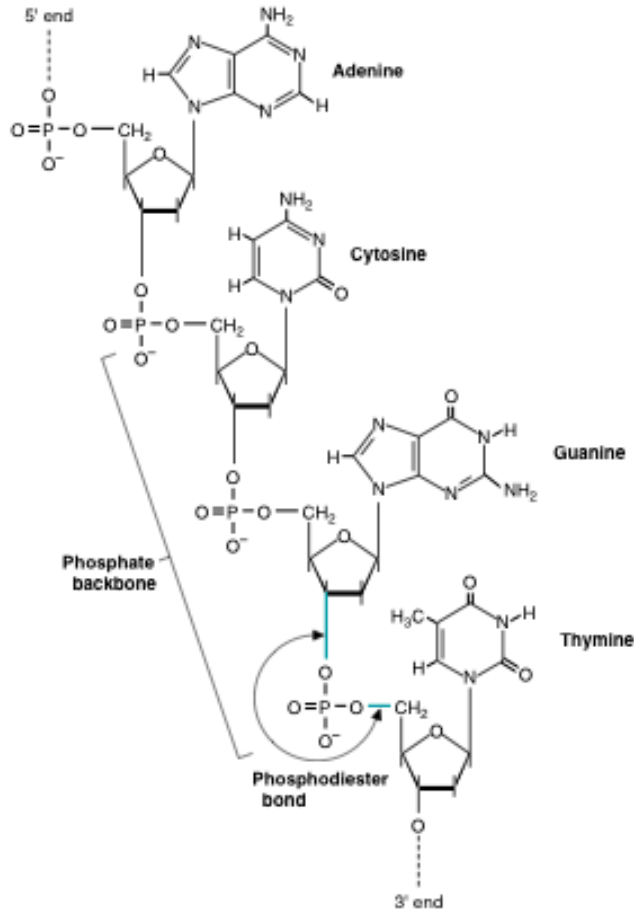


Figure showing the structure of a polynucleotide.

There are four designations used to describe the structure of DNA and RNA.

Primary structure: the sequence of nucleotides (bases) along the chain. The convention is to list the base order from 5' to 3'. Eg. 5' – AGGCTAAC – 3'

Secondary structure: the 3-dimensional structure of the backbone. Eg. Double helix of DNA

Tertiary structure: the super-coiling of the secondary structure

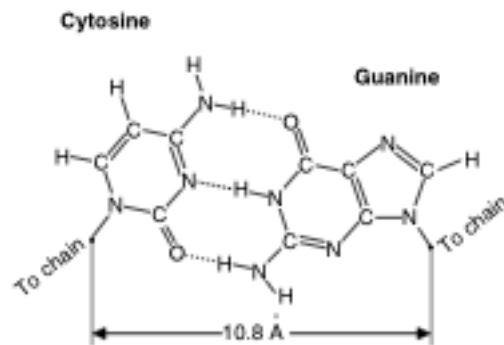
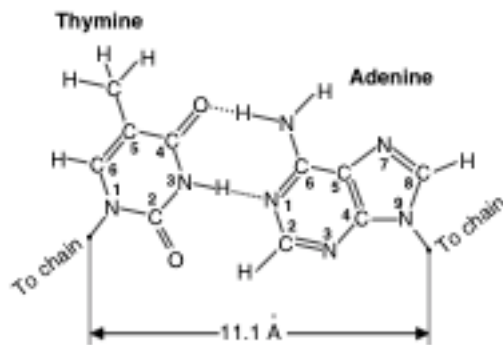
Quaternary structure: the interaction of nucleic acids with proteins

The DNA Double Helix:

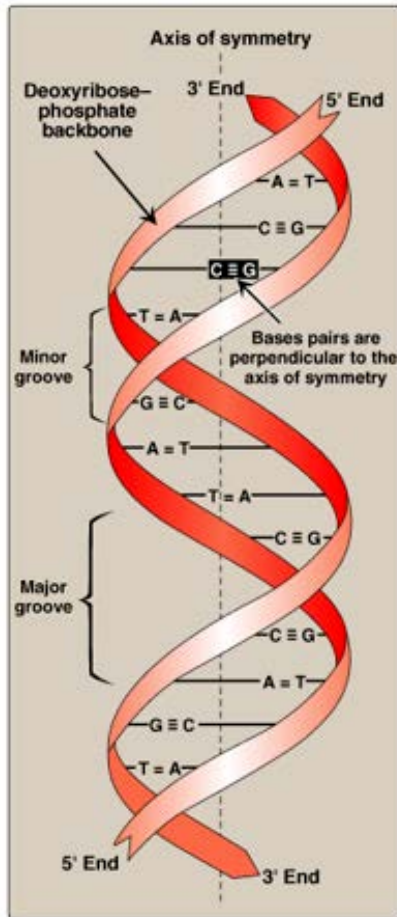
The structure of the DNA molecule, now referred to as B-type DNA was proposed by James Watson and Francis Crick in 1953. The model of the DNA structure was developed based on X-ray diffraction patterns produced by Rosalind Franklin and Maurice Wilkins, the base ratios of Erwin Chargaff, and biophysical data of DNA.

The work of Watson and Crick provided the following features of the DNA double helix:

- a. Two complementary anti-parallel strands coiled (right handed) along a common axis.
- b. The chains are held together by hydrogen bonds: A-T and G-C.
- c. Bases are on the inside with their planes being perpendicular to the helix axis.
- d. The sugar and phosphate groups are on the outside.
- e. The planes of the sugars are almost at right angles to those of the bases.
- f. The diameter of the helix is about 2.0 nm containing 10.4 base pairs per turn of the helix (3.4 nm).
- g. Stabilization of the helix is achieved by hydrogen bonds, stacking interactions (van der Waals forces) of the bases, hydrophobic effects and charge effects.
- h. The helix contains a major groove and a minor groove which provide sites for DNA-protein binding.



Hydrogen bonds the complimentary base pairs and thus the DNA helix itself together. The weakness of hydrogen bonding allows the strands of the DNA to separate during DNA replication and transcription.



A figure displaying the major structural features of the DNA double helix (B-form DNA) as outlined above.

Other forms of DNA:

There are a number of other forms of DNA structure that have been observed, however, it is uncertain what biological role (if any) these have.

A-type DNA: A-type structure is favoured when a DNA solution is dehydrated. Like B-DNA, A-DNA is a right handed double helix with 11 bp per turn and diameter of about 2.6 nm. The planes of the bases are not perpendicular like in B-DNA and as a result there is a deeper major groove and a shallower minor groove. This form of DNA has been observed in short DNA molecules, double-stranded RNA and DNA/RNA hybrids. It has not been observed in the cell.

Z-type DNA: Z-type structure is a left handed helix with 12 bp per turn. The backbone exhibits a zig-zag extended pattern. This structure is formed in sequences of alternating purine and pyrimidines. Eg. GC. There is evidence that this structure is present in both eukaryotes and prokaryotes, and it has been suggested to have a role in recombination and regulation of gene expression.

RNA:

RNA molecules are synthesized as single stranded ribonucleic acid polymers. Following synthesis, however, transfer RNA (tRNA) and ribosomal RNA (rRNA) fold and form intra-molecular hydrogen bonding that resulting to regions of double stranded RNA.

There are at least four different types of RNA:

mRNA (2% of total cellular RNA) – the message that directs the amino acid sequence of proteins

rRNA (82%) – a component of the structure of the ribosome

snRNA (less than 1%) – involved in mRNA splicing and gene regulation

tRNA (16%) – adaptor molecules that transport amino acids to the ribosomes

In tRNA, the intra-chain complementary base pairing produces a clover leaf shape with stem-loop or hairpin turns. The 3'- end containing the sequence CCA is the point of attachment of the amino acid (at the terminal A residue). An anticodon sequence (triplet) basepairs with a specific codon on the mRNA to facilitate polypeptide synthesis during translation.

Physicochemical properties of nucleic acids:

DNA denaturation: breaking of hydrogen bonds and unstacking of the bases resulting in the separation of strands. Denaturation *in vitro* can be accomplished by heating or exposure to alkali.

Renaturation: reassociation (annealing) of strands with slow cooling after denaturation. The rate of renaturation depends on the DNA concentration, time, and DNA complexity. Renaturation rate is inversely proportional to the complexity of the DNA.

Melting temperature T_m : the temperature at which double-stranded DNA is half-denatured. The melting temperature increases with increasing GC content.

Hybridization: the formation of hybrid DNA molecules or DNA/RNA hybrids between single stranded molecules containing certain amounts of complementarity. The hybridization process is the foundation for a number of molecular biology reactions and analysis.

Cyclic Nucleotides: cyclic AMP and cyclic GMP

3', 5'-cyclic derivatives of AMP and GMP are involved as 2nd messengers in the signal transduction from hormones to intracellular enzymes in eukaryotes. In prokaryotes, they are involved in intracellular signalling and gene expression.

Clinical Correlations – Nucleoside and Nucleotide analogs:

Synthetic analogs of bases, nucleosides and nucleotides are used in chemotherapy. The mechanism of action involves inhibition of enzymes that function in nucleic acid synthesis, or incorporation into DNA resulting in termination of DNA replication.

Nucleoside analogs:

Azidothymidine (AZT) and **Didanosine (ddl)** are both nucleoside analogs that both lack a free 3' hydroxyl (-OH) group. These drugs have high affinity for reverse transcriptase enzyme and inhibit their activity. They are therefore used in the treatment of retro-viral infections such as HIV.

Acyclovir possesses an open chain structure in place of the pentose sugar ring structure. Acyclovir is a potent inhibitor of viral DNA polymerase and causes termination of DNA replication. It is used in the treatment of Herpes virus.

Nucleotide analogs:

Tenofovir is a nucleotide analogs that possess an open chain structure in place of the pentose sugar ring structure. Tenofovir is a potent reverse transcriptase inhibitor that is used to treat HIV infection.

Arabinosides:

Cytosine arabinoside and **adenosine arabinoside** possess unusual planar structures and as a result inhibit the activity of DNA polymerase as well as inducing DNA damage by incorporation into a DNA during replication. These drugs are used as anti-cancer drugs by selectively destroying the rapidly dividing cancer cells.

Cytidine Analogs:

The cytidine analogs **5-aza-2'-deoxycytidine** and **5-azacytidine** are cytidine analogs that disrupt the epigenetic process of DNA methylation. These analogs possess a N atom at position 5 of cytidine and therefore prevent the DNA methyl-transferase enzymes from methylation cytosine at position 5. In addition, they have been found to be direct inhibitors of DNA methyl-transferase enzymes. DNA methylation is an important mechanism of gene regulation (transcription) in eukaryotes.

Analogs that disrupts other biochemical pathways (later lectures):

5-flourouracil is an anti-cancer agent. In cells it is converted to FdUMP which inhibits thymidylate kinase, and thus inhibiting synthesis of thymine nucleotides that are required for DNA synthesis.

Azathioprine is metabolized to 6-mercaptopurine which suppresses rejection during organ transplantation.

Allopurinol is a treatment for gout. It inhibits xanthine oxidase, thus lowering the conversion of purine bases to uric acid end products. The purines are excreted as xanthine and hypoxanthine which are more soluble than uric acid.

Adenosine (Adenocard) IV is used to treat supraventricular tachycardia. It slows the heart rate. It also plays a role in sleep regulation whereby high levels correspond to extended periods of wakefulness while low levels correspond to periods of sleep.